Environment-Responsive Coarse-Grained Bead Dynamics

VSEPR-SIM Project

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Abstract

The anisotropic surface-mapped bead, as developed in the companion document, is a self-contained physical object: it carries a descriptor, an orientation, and inertial properties, and it interacts with other beads through orientation-dependent harmonic-space potentials. However, its state is intrinsic—it has no awareness of its local environment. This document introduces the *environment-responsive* extension, in which each bead acquires a set of fast observables (local density, coordination number, orientational order) and a slow internal state variable with relaxation dynamics and memory. The coupling of this extended state back into the interaction kernels enables density-dependent behaviour, spontaneous ordering, hysteresis, and metastability. The result is a coarse-grained bead that does not merely exist in a simulation but *participates* in collective material behaviour.

1 Introduction and Motivation

The companion document *Anisotropic Surface-Mapped Coarse-Grained Beads for Molecular Systems* establishes the canonical bead state:

$$B = \left\{\mathbf{R}_B,\ \mathbf{Q}_B,\ \{c_{\ell m}^{(k)}\},\ M_B,\ Q_B^{\mathrm{tot}},\ \mathbf{I}_B\right\} \tag{1}$$

This state is sufficient for pairwise anisotropic interactions and rigid-body dynamics. However, it is *context-blind*: a bead in vacuum behaves identically to a bead in a dense crystal, because nothing in B encodes the local surroundings.

Real molecular systems exhibit environment-dependent behaviour at every scale:

- aromatic stacking stabilises in dense phases but not in dilute gas
- electrostatic screening increases with local coordination
- packing-induced rigidity modifies effective interaction strength
- interfacial beads experience asymmetric environments
- ordering transitions emerge from local alignment reinforcement

To capture these phenomena, the bead must acquire a second layer of state that responds to and encodes local structure. This document defines that layer, establishes its design constraints, and specifies its coupling to the existing interaction machinery.

2 Design Principles

The environment-responsive extension is governed by four non-negotiable rules. Violation of any one produces a model that is clever but useless.

2.1 Rule 1: X_B Is Derived, Not Invented

Every component of the environment state must be computable from measurable local structure—positions, orientations, and distances of neighbouring beads. Arbitrary labels, ad hoc classifications, and externally imposed phase tags are prohibited. The bead must *discover* its environment from its neighbours, not be *told* what environment it is in.

2.2 Rule 2: Separate Fast Observables from Slow State

The extension contains two distinct temporal classes:

- **Fast observables:** recomputed every dynamics step from the current neighbour configuration. These are instantaneous measurements, not state variables.
- **Slow state:** evolves via a relaxation equation with a finite timescale. This is the memory layer—it integrates fast observations over time.

Conflating these two classes destroys the model's ability to exhibit hysteresis and metastability.

2.3 Rule 3: X_B Does Not Replace the Descriptor

The environment state modifies *behaviour*, not *identity*. The descriptor coefficients $\{c_{\ell m}^{(k)}\}$ encode what the bead intrinsically is. The environment state X_B encodes what the bead is currently experiencing. These are distinct concepts with distinct update schedules, and the architecture must keep them separate.

2.4 Rule 4: Keep It Low-Dimensional

If X_B becomes a 20-variable state vector, the result is a worse atomistic model with extra steps. The environment extension must remain minimal: enough variables to capture the essential physics, and no more. Every additional variable must justify its existence by producing observable behaviour that cannot be obtained without it.

3 Extended Bead State

The environment-responsive bead is defined by the augmented state:

$$\boxed{\mathbb{B}_B = \{B,\ X_B\}} \tag{2}$$

where B is the canonical intrinsic state from the companion document and

$$X_B = \{\rho_B,\ C_B,\ P_{2,B},\ \eta_B\} \tag{3}$$

is the environment-responsive state.

The first three components (ρ_B, C_B, $P_{2,B}$) are fast observables—instantaneous measurements recomputed every step. The fourth component (η_B) is a slow internal state variable with relaxation dynamics and memory.

3.1 State Hierarchy

The full bead state now spans five layers:

Layer	Contents	Update	Role
A (Identity)	$\{c_{\ell m}^{(k)}\}$, M_B, Q_B^{tot}, $\mathbf{I}_B$	Fixed unless remapped	What the bead is
B (Pose)	$\mathbf{R}_B$, $\mathbf{Q}_B$	Every step	Where and how it is oriented
C (Resolution)	$\ell_{\max}$, channels, residuals	At adaptation	Whether the descriptor suffices
D (Provenance)	Fragment ID, frame method	At mapping only	Traceability
E (Environment)	ρ_B, C_B, $P_{2,B}$, η_B	**Every step**	**What the bead experiences**

Layer E is the new contribution of this document. It depends on Layer B (the bead needs to know where its neighbours are) but does not depend on Layer A (the environment state does not care about descriptor coefficients).

4 Fast Observables

Fast observables are instantaneous measurements of the bead's local environment, recomputed every dynamics step from neighbour positions and orientations. They are not state variables—they carry no memory and respond immediately to configuration changes.

4.1 Local Density: ρ_B

$$\rho_B = \sum_{C \neq B} w(r_{BC}) \tag{4}$$

where $r_{BC} = |\mathbf{R}_C - \mathbf{R}_B|$ and $w(r)$ is a smooth, normalised weighting function with compact support.

Physical meaning. ρ_B measures the degree of crowding around bead B. It is large in dense packings and small in dilute environments.

Weighting function. A Gaussian with smooth cutoff provides a suitable default:

$$w(r) = \begin{cases} \exp\left(-\dfrac{r^2}{2\sigma_\rho^2}\right) \cdot f_{\mathrm{sw}}(r;\, r_c,\, \delta) & r < r_c \\ 0 & r \geq r_c \end{cases} \tag{5}$$

where σ_ρ controls the spatial scale of the density measurement, r_c is the cutoff radius, and f_{sw} is a switching function ensuring continuity at r_c.

Gradient. The gradient of ρ_B with respect to $\mathbf{R}_B$ is:

$$\nabla_{\mathbf{R}_B} \rho_B = -\sum_{C \neq B} w'(r_{BC})\, \hat{r}_{BC} \tag{6}$$

This is needed if ρ_B couples into the potential energy, as the resulting force contribution must be included for energy conservation.

4.2 Coordination Number: C_B

$$C_B = \sum_{C \neq B} \mathbf{1}_{r_{BC} < r_c} \tag{7}$$

Physical meaning. C_B is the integer (or near-integer) count of neighbours within a cutoff distance. It provides a crude but extremely informative measure of neighbourhood topology.

Smoothed variant. For differentiability, a soft coordination number may be used:

$$C_B^{\text{soft}} = \sum_{C \neq B} \frac{1 - (r_{BC}/r_c)^n}{1 - (r_{BC}/r_c)^m} \tag{8}$$

with $m > n$ (e.g., $n = 6$, $m = 12$), following standard PLUMED-style switching functions.

4.3 Orientational Order Parameter: $P_{2,B}$

$$P_{2,B} = \frac{1}{N_B} \sum_{C \in \mathcal{N}_B} \frac{1}{2} \left(3 \left(\hat{n}_B \cdot \hat{n}_C \right)^2 - 1 \right) \tag{9}$$

where $\hat{n}_B$ is the primary orientation axis of bead B (e.g., the normal to an aromatic plane), $\mathcal{N}_B$ is the set of neighbours within cutoff, and $N_B = |\mathcal{N}_B|$.

Physical meaning. $P_{2,B}$ is the local second-rank Legendre order parameter. It detects the degree of orientational alignment between a bead and its neighbours:

- $P_{2,B} \approx 1$: perfect parallel alignment (nematic order, π-stacking)
- $P_{2,B} \approx 0$: random orientational distribution (isotropic liquid)
- $P_{2,B} \approx -\frac{1}{2}$: perfect perpendicular alignment (anti-nematic)

Gradient. The gradient with respect to orientation $\mathbf{Q}_B$ generates a torque contribution when $P_{2,B}$ couples into the potential. Explicit evaluation requires differentiation of $\hat{n}_B \cdot \hat{n}_C$ with respect to the orientation frame.

4.4 Observable Interpretation Table

Regime	ρ_B	C_B	$P_{2,B}$
Gas / dilute	low	low	≈ 0
Isotropic liquid	medium	medium	≈ 0
Aligned / crystal	high	high	large (> 0)
Interface	asymmetric	variable	unstable

These three observables jointly provide *phase awareness* without requiring explicit phase labels, external order parameters, or global thermodynamic state identification. The bead discovers its regime from local structure alone.

5 Slow Internal State Variable

The fast observables respond instantaneously to configuration changes. To capture memory, hysteresis, and metastability, the bead requires a *slow* state variable that integrates environmental information over time.

5.1 Definition

The slow internal state variable $\eta_B \in [0, 1]$ evolves according to a first-order relaxation equation:

$$\boxed{\frac{d\eta_B}{dt} = \frac{1}{\tau}\left(f(\rho_B,\, C_B,\, P_{2,B}) - \eta_B\right)} \tag{10}$$

where:

- $\tau > 0$ is the relaxation timescale (controls memory strength)
- $f(\rho_B, C_B, P_{2,B}) \in [0, 1]$ is the target function that maps instantaneous observables to a desired internal state
- η_B relaxes *toward* f but does not track it instantaneously

5.2 Physical Interpretation

η_B represents a scalar measure of the bead's integrated local environment. Depending on the system under study, it may be interpreted as:

- **stacking participation:** $\eta \approx 1$ when the bead has been in a well-ordered stacking environment for a sustained period
- **local ordering membership:** $\eta \approx 1$ in regions of persistent orientational alignment
- **crowding-induced rigidity:** η increases with sustained high density, reflecting local stiffening
- **phase-like scalar:** $\eta \approx 0$ in disordered regions, $\eta \approx 1$ in ordered regions

The interpretation is chosen *once* for a given model and held consistently. Attempting to encode multiple distinct physical phenomena into a single scalar produces confusion, not generality.

5.3 Relaxation Timescale

The parameter τ controls the temporal response:

- $\tau \to 0$**:** $\eta_B \to f$ instantaneously; no memory, no hysteresis
- τ **moderate:** η_B lags behind fast observables; transient memory, damped oscillation
- $\tau \to \infty$**:** η_B barely responds; strong memory, persistent metastability

The choice of τ is a physical modelling decision, not a numerical parameter. It sets the timescale over which the bead "remembers" its recent environment.

5.4 Minimal Target Function

The simplest viable target function is a linear combination of the fast observables:

$$f(\rho_B, C_B, P_{2,B}) = \alpha\,\hat{\rho}_B + \beta\,\hat{P}_{2,B} \tag{11}$$

where $\hat{\rho}_B$ and $\hat{P}_{2,B}$ are normalised to $[0, 1]$ and $\alpha + \beta = 1$.

With this choice, the evolution equation becomes:

$$\frac{d\eta_B}{dt} = \frac{1}{\tau}\Big(\alpha\,\hat{\rho}_B + \beta\,\hat{P}_{2,B} - \eta_B\Big) \tag{12}$$

This is the **minimal viable specification**: two parameters (α, β) control the relative importance of density and order, while τ controls memory depth. Three parameters total. If this does not produce interesting behaviour, the problem is not that more variables are needed.

5.5 Steady-State Analysis

At steady state ($d\eta_B/dt = 0$):

$$\eta_B^{\text{ss}} = f(\rho_B, C_B, P_{2,B}) \tag{13}$$

The slow variable converges to the target function. Out of steady state, η_B lags behind f, creating:

- **hysteresis:** η_B retains high values after the environment has become disordered, and vice versa
- **metastability:** beads in a locally ordered region resist disorder because their η_B is still high, reinforcing the interactions that maintain order
- **nucleation asymmetry:** ordering is harder to initiate (requires sustained $f > 0$) than to maintain (high η stabilises itself)

5.6 Numerical Integration

For small $\Delta t/\tau$, the evolution equation is integrated via forward Euler:

$$\eta_B(t + \Delta t) = \eta_B(t) + \frac{\Delta t}{\tau}\,(f_B(t) - \eta_B(t)) \tag{14}$$

For stiff cases ($\Delta t/\tau \gtrsim 0.5$), the implicit (exact) solution is preferred:

$$\eta_B(t + \Delta t) = f_B(t) + (\eta_B(t) - f_B(t))\,e^{-\Delta t/\tau} \tag{15}$$

Both forms are unconditionally stable for $\tau > 0$.

6 Coupling X_B into the Interaction Machinery

The environment state is physically meaningful only if it modifies the bead's interactions. Three coupling strategies are defined, ordered by increasing aggressiveness.

6.1 Option A: Kernel Modulation (Recommended)

The per-ℓ radial kernels acquire dependence on the internal state variables of the interacting beads:

$$K_k(\ell, r) \longrightarrow K_k(\ell, r;\, \eta_A, \eta_B) \tag{16}$$

The simplest modulation is multiplicative:

$$K_k(\ell, r;\, \eta_A, \eta_B) = K_k(\ell, r) \cdot g_k(\eta_A, \eta_B) \tag{17}$$

Channel-specific modulations.

- **Dispersion enhancement under ordering:**

$$K_{\text{disp}}(\ell, r;\, \eta_A, \eta_B) = K_{\text{disp}}(\ell, r) \cdot (1 + \gamma_{\text{disp}}\, \bar{\eta}_{AB}) \tag{18}$$

  where $\bar{\eta}_{AB} = \frac{1}{2}(\eta_A + \eta_B)$. This stabilises π-stacking when both beads are in an ordered environment.

- **Electrostatic screening under crowding:**

$$K_{\text{elec}}(\ell, r;\, \eta_A, \eta_B) = K_{\text{elec}}(\ell, r) \cdot (1 - \gamma_{\text{elec}}\, \bar{\eta}_{AB}) \tag{19}$$

  This damps electrostatic anisotropy in dense, disordered regions where screening is expected.

- **Steric hardening under compression:**

$$K_{\text{steric}}(\ell, r;\, \eta_A, \eta_B) = K_{\text{steric}}(\ell, r) \cdot (1 + \gamma_{\text{steric}}\, \bar{\eta}_{AB}) \tag{20}$$

  This increases effective excluded volume under sustained crowding.

Advantages. Kernel modulation is the cleanest coupling because:

- it preserves the structure of the harmonic-space interaction engine
- it introduces no new interaction terms
- all existing per-ℓ decompositions remain valid
- gradients with respect to η are straightforward

6.2 Option B: Descriptor Scaling

The descriptor coefficients themselves become functions of the internal state:

$$c_{\ell m}^{(k)} \longrightarrow c_{\ell m}^{(k)}(\eta_B) = c_{\ell m}^{(k)} \cdot h_k(\eta_B) \tag{21}$$

Examples:

- steric coefficients shrink under compression ($h_{\text{steric}} = 1 - \delta\, \eta_B$)
- electrostatic anisotropy weakens in dense regions ($h_{\text{elec}} = 1 - \delta\, \eta_B$)

Caution. Descriptor scaling is more aggressive than kernel modulation because it modifies the bead's *apparent identity* based on environment. This blurs the separation between identity (Layer A) and environment (Layer E). Use only when kernel modulation is insufficient.

6.3 Option C: Additive Environment Energy

A separate energy contribution is defined that depends directly on the environment observables:

$$U_{\text{env}} = \sum_B \Psi(\rho_B,\, C_B,\, P_{2,B},\, \eta_B) \tag{22}$$

This captures:

- packing penalties (high ρ_B increases energy)
- ordering rewards (high $P_{2,B}$ decreases energy)
- frustration (competing density and order terms)

Force contribution. The additive environment energy contributes to forces via:

$$\mathbf{F}_B^{\text{env}} = -\nabla_{\mathbf{R}_B} U_{\text{env}} = -\sum_B \frac{\partial \Psi}{\partial \rho_B} \nabla_{\mathbf{R}_B} \rho_B - \frac{\partial \Psi}{\partial P_{2,B}} \nabla_{\mathbf{R}_B} P_{2,B} - \cdots \tag{23}$$

This requires gradients of all fast observables with respect to bead positions and orientations.

6.4 Recommended Starting Configuration

The minimal viable coupling uses Option A (kernel modulation) only, with a single modulation parameter per channel:

$$K_k(\ell, r;\, \eta_A, \eta_B) = K_k(\ell, r) \cdot (1 + \gamma_k\, \bar{\eta}_{AB}) \tag{24}$$

Total new parameters: γ_{steric}, γ_{elec}, γ_{disp}, α, β, τ—six parameters beyond the existing interaction model. This is sufficient. If it does not produce interesting behaviour, the problem is not that more variables are needed.

7 Emergent Behaviour

The coupling of environment state into the interaction kernels enables qualitatively new phenomena that are impossible in a context-blind model.

7.1 Spontaneous Ordering

When $\gamma_{\text{disp}} > 0$, beads that are already in an ordered environment ($\eta \approx 1$) interact more strongly via the dispersion channel. This reinforces the ordering that produced high η in the first place, creating a positive feedback loop:

$$\text{high } P_{2,B} \;\rightarrow\; \text{high } f_B \;\rightarrow\; \text{high } \eta_B \;\rightarrow\; \text{stronger } K_{\text{disp}} \;\rightarrow\; \text{higher } P_{2,B} \tag{25}$$

This is the mechanism for spontaneous stacking, liquid-crystal alignment, and local crystallisation.

7.2 Density-Dependent Behaviour

Kernel modulation via ρ_B produces:

- steric hardening under compression
- electrostatic damping in dense environments (screening)
- dispersion saturation at high packing

These are physically expected behaviours that isotropic CG models achieve only through tabulated state-dependent potentials. Here they emerge from the local environment coupling.

7.3 Hysteresis and Metastability

Because η_B relaxes on a timescale τ, the system exhibits history-dependent behaviour:

- a bead that was recently in an ordered region retains high η even after the ordering dissolves, temporarily maintaining stronger interactions
- a bead in a disordered region requires sustained exposure to order before η rises enough to trigger reinforcement
- this asymmetry between ordering and disordering produces hysteresis in density-temperature phase behaviour

7.4 Interface Sensitivity

Beads at interfaces between ordered and disordered regions experience asymmetric environments:

- high $P_{2,B}$ on the ordered side, low on the disordered side
- intermediate η_B due to mixed f_B values
- the gradient in η across the interface produces a natural surface tension analogue without explicit interfacial terms

7.5 Summary of Accessible Phenomena

Phenomenon	**Mechanism**	**Key parameter**
Spontaneous stacking	$P_2 \to \eta \to K_{\text{disp}}$ feedback	γ_{disp}, β
Electrostatic screening	$\rho \to \eta \to K_{\text{elec}}$ damping	γ_{elec}, α
Steric hardening	$\rho \to \eta \to K_{\text{steric}}$ increase	γ_{steric}, α
Hysteresis	$\tau > 0$ delays η response	τ
Metastability	high η reinforces ordering	γ_{disp}, τ
Nucleation asymmetry	ordering requires sustained $f > 0$	τ, β
Interface sensitivity	$\nabla\eta$ across ordered/disordered boundary	all

8 Formal Specification Summary

8.1 Extended Bead State

$$\mathbb{B}_B = \{\underbrace{\mathbf{R}_B,\ \mathbf{Q}_B,\ \{c_{\ell m}^{(k)}\},\ M_B,\ Q_B^{\text{tot}},\ \mathbf{I}_B}_{B\ \text{(intrinsic)}},\ \underbrace{\rho_B,\ C_B,\ P_{2,B},\ \eta_B}_{X_B\ \text{(environment)}}\} \tag{26}$$

8.2 Fast Observable Definitions

$$\rho_B = \sum_{C \neq B} w(r_{BC}) \tag{27}$$

$$C_B = \sum_{C \neq B} \mathbf{1}_{r_{BC} < r_c} \tag{28}$$

$$P_{2,B} = \frac{1}{N_B} \sum_{C \in \mathcal{N}_B} \frac{1}{2} \left(3(\hat{n}_B \cdot \hat{n}_C)^2 - 1 \right) \tag{29}$$

8.3 Slow State Evolution

$$\frac{d\eta_B}{dt} = \frac{1}{\tau} \Big(\alpha \, \hat{\rho}_B + \beta \, \hat{P}_{2,B} - \eta_B \Big) \tag{30}$$

8.4 Kernel Modulation

$$K_k(\ell, r;\, \eta_A, \eta_B) = K_k(\ell, r) \cdot (1 + \gamma_k \, \bar{\eta}_{AB}) \tag{31}$$

8.5 Parameter Set

Parameter	Role	Typical range
α	density weight in target function	$[0, 1]$
β	orientational order weight	$[0, 1]$, $\alpha + \beta = 1$
τ	relaxation timescale	10–10^4 fs
γ_{steric}	steric kernel modulation	$[0, 1]$
γ_{elec}	electrostatic kernel modulation	$[-1, 0]$
γ_{disp}	dispersion kernel modulation	$[0, 2]$

8.6 Invariants

- $\eta_B \in [0, 1]$ at all times (clamp if target function overshoots)
- $|1 + \gamma_k \bar{\eta}| > 0$ for all channels (kernel must not change sign)
- Fast observables are recomputed before η is updated
- η is updated before interaction evaluation
- Energy conservation requires force contributions from $\partial K / \partial \eta \cdot \partial \eta / \partial \mathbf{R}$ when η is pose-dependent

9 Relationship to Companion Document

This document extends the companion *Anisotropic Surface-Mapped Coarse-Grained Beads for Molecular Systems* without modifying it.

Companion document	This document
Defines B (intrinsic state)	Defines X_B (environment state)
Pairwise interaction engine	Environment-modulated kernels
Static descriptor coefficients	Coefficients optionally scaled
No environment awareness	Phase awareness from local observables
Rigid-body dynamics only	Slow internal state with memory
Deterministic mapping	Deterministic observables + relaxation

The interface is clean: this document adds Layer E to the existing four-layer architecture and modulates the existing interaction kernels via multiplicative coupling. No existing equations are changed; the extension is purely additive.

10 Conclusion

The environment-responsive extension transforms the anisotropic bead from a static descriptor carrier into a material participant. By introducing three fast observables and one slow internal state variable, the bead acquires phase awareness, memory, and history-dependent interaction strength—all from six additional parameters.

The distinction is between:

- "the model works" — beads interact anisotropically and move under force and torque
- "the system behaves" — beads spontaneously order, resist disordering, screen electrostatics in dense regions, and exhibit hysteresis across phase boundaries

The first capability was established in the companion document. The second is established here.